

Report on the `racomDataset12class` Dataset

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Abstract

This report provides a detailed description of the `racomDataset12class`, a large-scale, synthetic dataset designed for machine learning applications, specifically for modulation classification. The provided scripts can generate two distinct versions of the dataset: a **Small Dataset** with 384,000 total samples and a **Large Dataset** with 960,000 total samples. Each sample is a Time-Frequency Distribution (TFD) image generated from 12 types of simulated communication and radar signals. This document outlines the dataset's structure, the complete data generation pipeline, signal configurations, channel modeling, and the final image representation for both versions of the dataset.

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1 Overall Dataset Description

The `racomDataset12class` is a synthetic dataset created for developing and evaluating machine learning models for modulation classification. The generation scripts are configured to produce two versions based on the number of samples per class. Each sample is a **224x224 pixel grayscale PNG image** that represents the Time-Frequency Distribution (TFD) of a simulated signal passed through a realistic channel model.

The dataset features 12 distinct signal modulation types, categorized as follows:

- **8 Communication Modulation Types:** A combination of digital and analog schemes.
- **4 Radar Waveform Types:** Common radar pulse modulations.

1.1 Dataset Versions and Sizes

Two versions of the dataset can be generated by modifying a parameter in the generation script.

- **[Small Dataset]**
 - **Samples per Class:** 32,000
 - **Total Samples:** 384,000
 - **Generation Note:** This dataset is generated by setting the `numFramesPerModType` variable to 4000 in the `mainProgramDataGeneration.m` script. The total of 32,000 samples per class is reached by generating 4,000 frames for each of the 8 SNR levels ($4,000 \times 8 = 32,000$).
- **[Large Dataset]**
 - **Samples per Class:** 80,000
 - **Total Samples:** 960,000
 - **Generation Note:** This dataset is generated by setting the `numFramesPerModType` variable to 10000. The total of 80,000 samples per class is reached by generating 10,000 frames for each of the 8 SNR levels ($10,000 \times 8 = 80,000$).

2 Dataset Structure and Organization

The dataset is organized in a hierarchical folder structure. It is recommended to use distinct root directory names for each version (e.g., `racomDataset12class_small`).

2.1 Directory Structure

- **Root Directory:** e.g., `racomDataset12class_small` or `racomDataset12class_large`.
- **Subdirectories:** The root folder contains 12 subdirectories, one for each modulation type.
 - **Communication:** BPSK, QPSK, GFSK, CPFSK, PAM4, 16-QAM, B-FM, DSB-AM.
 - **Radar:** LFM, Rect, Barker, StepFM.

2.2 File Naming Convention

Each image file follows a structured naming scheme that encodes its generation parameters:

`signal_snr<XX>_doppler<XXX>_img<XXXXXX>.png`

- **<XX>**: The two-digit index of the SNR level (from 01 to 08).
- **<XXX>**: The three-digit value of the Doppler shift in Hz (e.g., 004, 070).
- **<XXXXXX>**: The five-digit index of the image (e.g., up to 04000 for the small dataset, or up to 10000 for the large dataset).

3 Data Generation Pipeline

The generation process involves three main stages: baseband signal generation, channel simulation, and time-frequency image conversion.

3.1 Part A: Communication Signal Generation

- **Small Dataset**: $8 \text{ types} \times 32,000 \text{ samples/type} = \mathbf{256,000 \text{ images}}$.
- **Large Dataset**: $8 \text{ types} \times 80,000 \text{ samples/type} = \mathbf{640,000 \text{ images}}$.

3.1.1 Signal and Modulator Configurations

- **General Parameters**:
 - Sampling Rate (f_s): 200 kHz.
 - Samples per Symbol (sps): 8.
 - Samples per Frame (spf): 1024.
- **Source Data**:
 - Digital Modulations: Randomly generated integer data streams.
 - Analog Modulations (B-FM, DSB-AM): Audio from the `audio_mix_441.wav` file.
- **Specific Modulator Details**:
 - **Pulse-Shaped Modulations** (BPSK, QPSK, PAM4, 16-QAM): Use a root-raised cosine (RRC) filter for pulse shaping with a roll-off factor of 0.35 and a span of 4 symbols.
 - **Continuous Phase Modulations**:
 - * GFSK: Uses a Gaussian frequency pulse with a Bandwidth-Time (BT) product of 0.35 and a modulation index of 1.
 - * CPFSK: Uses a modulation index of 0.5.
 - **Analog Modulations**:
 - * B-FM: Simulates broadcast FM with a frequency deviation of 75 kHz and a 75 μ s pre-emphasis filter.
 - * DSB-AM: Modulates the audio onto a 50 kHz intermediate frequency (IF) carrier.

3.1.2 Channel Model

Each signal is impaired by a channel model that applies the following effects in sequence:

1. **Rician Multipath Fading:** A 7-path fading model is used.
 - Path Delays: $[0, 30, 150, 310, 370, 710, 1090]$ ns.
 - Average Path Gains: $[0, -1.5, -1.4, -3.6, -0.6, -9.1, -7.0]$ dB.
 - K-Factor: 4.
2. **Doppler Shift:** Randomly selected from the set $\{0, 4, 70, 300\}$ Hz for each signal.
3. **Clock Offset:** A random offset between -5 and +5 ppm is applied, causing frequency and timing drift. The center frequency for this calculation is 900 MHz (digital) or 100 MHz (analog).
4. **Additive White Gaussian Noise (AWGN):** Noise is added to achieve one of 8 SNR levels: $\{-5, 0, 5, 10, 15, 20, 25, 30\}$ dB.

3.2 Part B: Radar Signal Generation

- **Small Dataset:** $4 \text{ types} \times 32,000 \text{ samples/type} = \mathbf{128,000 \text{ images}}$.
- **Large Dataset:** $4 \text{ types} \times 80,000 \text{ samples/type} = \mathbf{320,000 \text{ images}}$.

3.2.1 Signal and Waveform Configurations

- **General Parameters:**
 - Sampling Rate (F_s): 100 MHz.
- **Randomized Waveform Parameters:** Each radar waveform is generated with unique, randomized parameters.
 - **Rect:** Pulse width is randomized.
 - **LFM:** Sweep bandwidth, pulse width, and sweep direction ('Up'/'Down') are randomized.
 - **Barker:** Number of chips (from $\{3, 4, 5, 7, 11\}$) and chip width are randomized.
 - **StepFM:** Number of steps (from $\{3, 4, 5, 7, 11\}$), frequency step size, and pulse width are randomized.

3.2.2 Channel Model

Radar signals are impaired in the following sequence:

1. **AWGN:** Noise is added first to achieve the target SNR from the set $\{-5, 0, 5, \dots, 30\}$ dB.
2. **Frequency Offset:** A deterministic frequency offset is applied.
3. **Rician Fading with Doppler:** The signal is passed through a channel with the same Rician parameters as the communication channel. The Doppler shift is also randomly selected from $\{0, 4, 70, 300\}$ Hz.

4 Final Image Representation

The final step for all signals is the conversion from a 1D time-series to a 2D image.

- **Framing:** The received signal is segmented into 1024-sample frames, and each frame's energy is normalized to unity.
- **Time-Frequency Transformation:** The **Smoothed Pseudo Wigner-Ville Distribution (SPWVD)** is calculated for each frame to provide a high-resolution view of the signal's energy across both time and frequency.
- **Image Properties:**
 - **Smoothing Filter:** A Kaiser window (`kaiser(101,20)`) is used.
 - **Resolution:** The final TFD is rendered as a **224x224** pixel array.
 - **Format:** The pixel values are rescaled, and the array is saved as a grayscale **PNG image**.